

Highlights

An allosteric metabolite glue couples AICAR inversely to one-carbon availability: a theoretical prediction and its therapeutic corollaries

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- Glue-regulated PPAT replaces demand-slaved purine synthesis in a formate model
- AICAR falls as one-carbon availability rises — an inverse coupling
- The inverse coupling needs the PPAT–NUDT5 glue; NUDT5 loss reverses its sign
- Calibration to binding data alone reproduces AICAR and PRPP data
- Corollaries follow for AMPK signalling and thiopurine cytotoxicity